Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/25/2004
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Project Title:	Confocal microscopy of the colloidal glass transition		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Task Description:	We study concentrated colloidal suspensions, a model system which has a glass transition. By using a confocal microscope to follow the three-dimensional motion of colloidal particles, we can directly observe the microscopic behavior responsible for the macroscopic viscosity divergence of glasses. To use direct imaging to quantify the microscopic behavior of particles as the colloidal glass transition is approached. Specifically, both structural properties and dynamical properties are to be measured and compared with theoretical descriptions of the glass transition. We also seek to characterize the aging of colloidal glasses; we wish to determine how colloidal glasses change over time. The overarching goal is to clarify the mechanisms underlying the glass transition and to provide key data to test various theories of the glass transition. Confocal microscopy of the colloidal suspensions allows us to both directly follow the motion of individual particles and to visualize structural arrangements as the transition is approached. These		

measurements are crucial to testing and extending theoretical understanding of both dense colloidal suspensions and glasses in general. Moreover, the insight from these experiments will be useful to the processing of industrially important materials. Finally, by understanding how the glass transition occurs in these model systems and how the dynamics are affected by the system details, we may be able to design novel materials with specific properties.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

In this second year of funding, we continued our experiments perturbing dense colloidal suspensions with magnetic beads. The work was led by postdoctoral fellow Piotr Habdas, and conducted with three graduate students: Rachel Courtland, David Schaar, and Shanshan Wu. Additional work has been done by Doug Anderson (who graduated from Emory University in May 2003 with a B.S. degree in physics), and by Amy Franciscovich (an Emory undergraduate). Dense colloidal suspensions are a simple model system which becomes glassy when the colloidal particle concentration is above a critical threshold. Below this threshold, the sample is analogous to a liquid, and colloidal particles move around the sample due to Brownian motion. The glass transition is identified at the point where the diffusion of colloidal particles through the sample can no longer be observed on reasonable experimental time scales. The goal of our experiments is to observe how such samples respond to being locally perturbed in a variety of ways. In particular, we wish to characterize how the response changes in liquid-like samples as the transition is approached.

Much of the work has focused on the drag force exerted by the colloidal suspension on an isolated magnetic bead. The magnetic beads are 5 mm in diameter, and the colloidal particles are 2 mm in diameter. We find that the drag force on a magnetic bead depends on the pulling speed in an unusual manner. Below a threshold force, the magnetic bead is nearly motionless, and moves only due to Brownian motion. Above this force, the bead is pulled with a fluctuating velocity. The relationship between force and velocity becomes increasingly nonlinear as the colloidal glass transition is approached. However, the velocity fluctuations do not change in magnitude or character near the glass transition. These results have been accepted for publication in Europhysics Letters.

We are also carefully investigating the motion of the colloidal particles surrounding the magnetic bead; this was work started by Shanshan Wu and currently continued by Piotr Habdas. As the magnetic bead moves, nearby colloidal particles must move out of the way. The motion is correlated over a length scale of ~2-4 particle diameters away from the magnetic bead. We are conducting experiments to see how this length scale changes as the glass transition approaches, and have preliminary evidence suggesting that it increases. If so, this would be an exciting connection to the fundamental problem of the glass transition. As a sample becomes more glassy, its viscosity rises dramatically by many orders of magnitude. If this viscosity change is associated with our length scale change, it might help us understand how the viscosity is linked with structural properties of the sample. Our preliminary data resulted in Shanshan Wu's master's thesis (July 2003).

We have also conducted magnetic bead experiments in several new areas.

* Doug Anderson is studying what happens when the force applied to the magnetic bead is removed and reapplied. The force duration is short enough that permanently rearrangements do not occur; however, the surrounding colloidal particles do move slightly and then return to their original locations. The key question is whether each time, the same particles move in response to the applied force, or if it is different particles. This will reveal if there are inherent structural features which the magnetic bead probes – for example, suggesting that one could predict them in advance, or try to relate them to the motion of the magnetic probe.

* Rachel Courtland is studying the behavior of a magnetic bead that is slowly oscillated in a circle. Because the forcing frequency is known, Fourier techniques are proving useful in the data analysis. An additional advantage of this technique is that the sample is not permanently disturbed. (In the translation experiments, the magnetic bead is forced to move through the sample and causes permanent rearrangements. We are looking to see if the behavior is different in the oscillatory experiments, where particles are only slightly perturbed from their equilibrium position.)

* Amy Franciscovich is studying how the force/velocity relationship changes when the size ratio between the magnetic bead and the colloidal particles is varied. Our preliminary data suggests that we see similar force/velocity behavior, that is, a threshold force for motion and a nonlinear drag force above this threshold.

* David Schaar is studying how the sample responds when the applied force is below the threshold force for motion. He finds that the sample behaves like a Hookean spring, with a magnetic bead displacement proportional to the exerted force ($F = k x$). We are carefully characterizing the properties of the local effective spring constant k, which seems to vary from location to location within a single sample. The results are in reasonable agreement with theoretical predictions made by K. Schweizer. A portion of this experimental work resulted in Schaar's master's thesis (October 2003).

An undergraduate (Ted Brzinski) has been studying hydrogel particles, which change their size when the temperature is changed. We are hoping to use these particles as a way to study the glass transition in a controlled fashion, i.e., to carefully tune the system to the glass transition packing fraction by controlling the temperature. These experiments may help us study more closely the behavior as we get asymptotically close to the glass transition; our previous work has been somewhat limited by the difficulty in selecting the packing fraction. In this past year, we've found difficulties with crystallization of the samples. The crystallization problem at the moment seems to be solved by using particles which are more polydisperse.

Christina Wallace (an undergraduate) and Gianguido Cianci (a graduate student) have begun studies of the aging of colloidal glasses. By stirring a dense sample vigorously, the sample is initialized, and the subsequent dynamics show aging: the time scale for particle motion grows with the age of the sample (the time since stirring ended). In our first year of funding, we studied the aging in monodisperse colloidal samples. In this second year, we've been focusing on binary colloidal suspensions.

These experiments are motivated in part because of our planned flight experiment, as part of PCS-3. We have provided an engineering test sample to NASA/Glenn and are continuing to work with them to determine ways to load the very dense samples into the PCS-3 apparatus. We have also done some testing of our samples in collaboration with Luca Cipelletti, using a light-scattering apparatus in Dr. Cipelletti's lab in Montpellier. Rachel Courtland went to Montpellier for two weeks in June 2003 to conduct these tests. At this point, it is uncertain if the PCS-3 experiments will be able to

	be conducted, given changing plans for future NASA flights.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Courtland, R.E.; Weeks, E.R. "Direct visualization of ageing in colloidal glasses" J. Phys: Cond. Mat. 15, S359-S365 , Jan-2003
Articles in Peer-reviewed Journals	Habdas, P.; Schaar, D.; Levitt, A.C.; Weeks, E.R. "Forced motion of a probe particle near the colloidal glass transition" Accepted for publication in Europhys. Lett. , May-2004
Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" American Chemical Society, invited talk Sep-2003
Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" University of Deleware, solid state physics seminar Nov-2003
Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" Yale University, mechanical engineering seminar Oct-2003
Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" University of Illinois, materials science colloquium Oct-2003
Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" University of Michigan, seminar Sep-2003
Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" Georgia Tech, mechanical engineering seminar Sep-2003
Significant Media Coverage	Covington M. "The feel of food fascinates Emory researcher. Article on PI's research." 90.1 WABE, Atlanta NPR station, January 27, 2004. http://wabe.org/post/feel-food-fascinates-emory-researcher ; accessed 5/25/2004., Jan-2004